

1. Administrative & Study Identification

Full Study Title: A Study of Nivolumab in Combination With Ipilimumab in Participants With Advanced Hepatocellular Carcinoma
Short Title/Acronym: CheckMate 9DW **Protocol Number:** CA209-9DW **NCT Number:** NCT04039607 **Trial Status:** Active, not recruiting **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab (Opdivo) and Ipilimumab (Yervoy) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To compare the overall survival (OS) of nivolumab plus ipilimumab versus standard of care (SOC) (sorafenib or lenvatinib) in all randomized participants with advanced hepatocellular carcinoma (HCC) who have not received prior systemic therapy. **Secondary Objectives:** To evaluate overall response rate (ORR), duration of response (DOR), time to symptom deterioration (TTSD), and to assess the safety profile of the combination.

2.2 Background & Rationale To address the need for a more effective first-line standard of care providing durable survival outcomes in patients with unresectable or metastatic hepatocellular carcinoma (uHCC). The study assesses whether the dual immune checkpoint inhibitor combination of nivolumab (PD-1 inhibitor) and ipilimumab (CTLA-4 inhibitor) is superior in efficacy compared to widely used tyrosine kinase inhibitors.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Lenvatinib or Sorafenib) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration **Target N\$:** 732 adults **Number of Sites:** Multicenter global study **Estimated Study Start:** 06-Jan-2020 **Estimated Primary Completion:** 08-Nov-2021 (with survival follow-up and primary analysis reported in 2024-2025)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed diagnosis of advanced or unresectable/metastatic hepatocellular carcinoma (HCC). 2. Child-Pugh score 5 or 6 and ECOG Performance Status 0

or 1. 3. No prior systemic therapy for advanced disease with at least one RECIST 1.1 measurable previously untreated lesion.

4.2 Exclusion Criteria 1. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 2. Prior liver transplant or episodes of hepatic encephalopathy ($\geq$ Grade 2) within 12 months prior to randomization. 3. Active brain metastases or leptomeningeal metastases.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 1 mg/kg of IV nivolumab and 3 mg/kg of IV ipilimumab every 3 weeks for a maximum of 4 doses, followed by 480 mg of IV nivolumab every 4 weeks. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms and treatment-related adverse events. **Prohibited:** Concurrent administration of other anti-cancer therapies or systemic immunosuppressive medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Scans, Labs
Baseline	Day 0	Randomization, First Dose	Interim
Week 3	$\pm$ 3	Safety Labs, Next Dose, Efficacy Assessment (Scans every 8-12 weeks)	
End of Study	Variable	Final Safety Assessment, End of Treatment Scans, Survival Follow-Up	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS), defined as the time from randomization to death due to any cause. **Measurement Method:** Kaplan-Meier method, log-rank tests, and Hazard Ratios (HR).

7.2 Secondary & Exploratory Endpoints Objective Response Rate (ORR) assessed by Blinded Independent Central Review (BICR) using RECIST 1.1. Duration of Response (DOR), Time to Symptom Deterioration (TTSD), and progression-free survival (PFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical evaluations, focusing specifically on immune-mediated AEs (e.g., rash, pruritus, fatigue, diarrhea, immune-mediated hepatitis, and gastrointestinal hemorrhage). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes (OS); Safety Set for all patients receiving at least one dose of the study medication. **Sample Size Justification:** $N=732$ adequately powered to detect a statistically significant improvement in median OS between the dual immunotherapy arm and the TKI comparator arm. **Handling of Missing Data:** Right-censoring techniques applied to patients alive at the data cutoff date or lost to follow-up for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:** eCRF tracking, comprehensive audit trails, and centralized data validation.

11. Study Identifiers & Governance

Primary ID: CA209-9DW **Posting ID:** 183936677833392508 **Registry Number:** NCT04039607 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 9DW **Official Regulatory Title:** A Randomized, Multi-center, Phase 3 Study of Nivolumab in Combination With Ipilimumab Compared to Sorafenib or Lenvatinib as First-Line Treatment in Participants With Advanced Hepatocellular Carcinoma

12. Clinical Framework (HCC Specific)

Primary Condition: Advanced Hepatocellular Carcinoma (uHCC) **Diagnosis Threshold:** Histologically confirmed, unresectable or metastatic, Child-Pugh score 5 or 6 **Medical Methodology:** Dual Immune Checkpoint Inhibitor Therapy (PD-1 + CTLA-4) **Optimization Target:** Extended Overall Survival (OS) and objective response achievement

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for first-line advanced HCC **Education Protocol (HCP):** Site initiation and comprehensive training on the recognition and management of immune-mediated adverse events (irAEs) **Education Protocol (Patient):** Informed consent process, symptom tracking, and self-reporting education for irAEs **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for objective radiographic tumor assessments

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity, followed by survival follow-up **Onsite Visit Cadence:** Every 3 weeks (first 4 doses), then every 4 weeks **Remote Monitoring Frequency:** Continuous monitoring and periodic survival status checks post-treatment **Checklist Review Interval:** Radiographic tumor evaluations every 8-12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				